	Caso	Intervalo de Confianza	Contraste de Hipótesis	Estadístico del Contraste	Región de rechazo
1	Para μ de $N\left(\mu,\sigma^2\right)$, σ^2 conocida	$\bar{x} \pm z_{\alpha/2}\dfrac{\sigma}{\sqrt{n}}$	$H_0: \mu = \mu_0;\ H_1: \mu \neq \mu_0$ $H_0: \mu \leq \mu_0;\ H_1: \mu > \mu_0$ $H_0: \mu \geq \mu_0;\ H_1: \mu < \mu_0$	$z_c = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \sim N\left(0,1\right)$	$\left\{\lvert z_c\rvert > z_{\alpha/2}\right\}$ $\left\{z_c > z_\alpha\right\}$ $\left\{z_c < -z_\alpha\right\}$
2	Para μ de $N\left(\mu,\sigma^2\right)$, σ^2 desconocida	$\bar{x} \pm t_{n-1,\alpha/2}\dfrac{s}{\sqrt{n}}$	$H_0: \mu = \mu_0;\ H_1: \mu \neq \mu_0$ $H_0: \mu \leq \mu_0;\ H_1: \mu > \mu_0$ $H_0: \mu \geq \mu_0;\ H_1: \mu < \mu_0$	$t_c = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$	$\left\{\lvert t_c\rvert > t_{n-1,\alpha/2}\right\}$ $\left\{t_c > t_{n-1,\alpha}\right\}$ $\left\{t_c < -t_{n-1,\alpha}\right\}$
3	Para μ caso general, n grande	$\bar{x} \pm z_{\alpha/2}\dfrac{s}{\sqrt{n}}$	$H_0: \mu = \mu_0;\ H_1: \mu \neq \mu_0$ $H_0: \mu \leq \mu_0;\ H_1: \mu > \mu_0$ $H_0: \mu \geq \mu_0;\ H_1: \mu < \mu_0$	$z_c = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim N\left(0,1\right)$	$\left\{\lvert z_c\rvert > z_{\alpha/2}\right\}$ $\left\{z_c > z_\alpha\right\}$ $\left\{z_c < -z_\alpha\right\}$
4	Para una proporción, p, caso general (n grande)	$\hat{p} \pm z_{\alpha/2}\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$	$H_0: p = p_0;\ H_1: p \neq p_0$ $H_0: p \leq p_0;\ H_1: p > p_0$ $H_0: p \geq p_0;\ H_1: p < p_0$	$z_c = \dfrac{\left(\hat{p} - p_0\right)}{\sqrt{\hat{p}\hat{q}/n}} \sim N\left(0,1\right)$	$\left\{\lvert z_c\rvert > z_{\alpha/2}\right\}$ $\left\{z_c > z_\alpha\right\}$ $\left\{z_c < -z_\alpha\right\}$
5	Para σ^2 de $N\left(\mu,\sigma^2\right)$	$\left(\dfrac{\left(n-1\right)s^2}{\chi^2_{n-1,\alpha/2}}, \dfrac{\left(n-1\right)s^2}{\chi^2_{n-1,1-\alpha/2}}\right)$	$H_0: \sigma^2 = \sigma_0^2;\ H_1: \sigma^2 \neq \sigma_0^2$ $H_0: \sigma^2 \leq \sigma_0^2;\ H_1: \sigma^2 > \sigma_0^2$ $H_0: \sigma^2 \geq \sigma_0^2;\ H_1: \sigma^2 < \sigma_0^2$	$\chi^2_c = \dfrac{\left(n-1\right)s^2}{\sigma_0^2} \sim \chi^2_{n-1}$	$\left\{\chi^2_c \notin \left[\chi^2_{n-1,1-\alpha/2}, \chi^2_{n-1,\alpha/2}\right]\right\}$ $\left\{\chi^2_c > \chi^2_{n-1,\alpha}\right\}$ $\left\{\chi^2_c < \chi^2_{n-1,1-\alpha}\right\}$

Nota:

$$s_p^2 = \frac{\left(n_1-1\right)s_1^2 + \left(n_2-1\right)s_2^2}{n_1+n_2-2}$$

$$f = \frac{\left(s_1^2/n_1 + s_2^2/n_2\right)^2}{\left(s_1^2/n_1\right)^2/(n_1-1) + \left(s_2^2/n_2\right)^2/(n_2-1)}$$

Se elige el entero más próximo a esta cantidad.

NOTA: Recuerda que usamos la notación t_{n-1; α} para referirnos al cuantil de la distribución t de Student con n-1 grados de libertad y una probabilidad de α a su derecha. Es decir, t_{n-1; α} es el valor tal que P(t > t_{n-1; α}) = α o P(t < t_{n-1; α}) = 1- α.

Caso		Intervalo de Confianza	Contraste de Hipótesis	Estadístico del Contraste	Región de rechazo
6	Para $\mu_1-\mu_2$ de $N\left(\mu_1,\sigma_1^2\right)$ y $N\left(\mu_2,\sigma_2^2\right)$ σ_1^2,σ_2^2 conocidas	$\left(\bar{x}-\bar{y}\right)\pm z_{\alpha/2}\sqrt{\sigma_1^2/n_1+\sigma_2^2/n_2}$	$H_0:\mu_1-\mu_2=d_0;\ H_1:\mu_1-\mu_2\neq d_0$ $H_0:\mu_1-\mu_2\leq d_0;\ H_1:\mu_1-\mu_2>d_0$ $H_0:\mu_1-\mu_2\geq d_0;\ H_1:\mu_1-\mu_2<d_0$	$z_c=\dfrac{\left(\bar{x}_1-\bar{x}_2\right)-d_0}{\sqrt{\sigma_1^2/n_1+\sigma_2^2/n_2}}\sim N\left(0,1\right)$	$\left\{\left\lvert z_c\right\rvert>z_{\alpha/2}\right\}$ $\left\{z_c>z_\alpha\right\}$ $\left\{z_c<-z_\alpha\right\}$
7	Para $\mu_1-\mu_2$ de $N\left(\mu_1,\sigma_1^2\right)$ y $N\left(\mu_2,\sigma_2^2\right)$, σ_1^2,σ_2^2 desconocidas, $\sigma_1^2=\sigma_2^2$	$\left(\bar{x}-\bar{y}\right)\pm t_{(n_1+n_2-2),\alpha/2}\,s_p\sqrt{1/n_1+1/n_2}$	$H_0:\mu_1-\mu_2=d_0;\ H_1:\mu_1-\mu_2\neq d_0$ $H_0:\mu_1-\mu_2\leq d_0;\ H_1:\mu_1-\mu_2>d_0$ $H_0:\mu_1-\mu_2\geq d_0;\ H_1:\mu_1-\mu_2<d_0$	$t_c=\dfrac{\left(\bar{x}_1-\bar{x}_2\right)-d_0}{s_p\cdot\sqrt{1/n_1+1/n_2}}\sim t_{n_1+n_2-2}$	$\left\{\left\lvert t_c\right\rvert>t_{n_1+n_2-2,\alpha/2}\right\}$ $\left\{t_c>t_{n_1+n_2-2,\alpha}\right\}$ $\left\{t_c<-t_{n_1+n_2-2,\alpha}\right\}$
8	Para $\mu_1-\mu_2$ de $N\left(\mu_1,\sigma_1^2\right)$ y $N\left(\mu_2,\sigma_2^2\right)$, σ_1^2,σ_2^2 desconocidas, $\sigma_1^2\neq\sigma_2^2$	$\left(\bar{x}-\bar{y}\right)\pm t_{f,\alpha/2}\sqrt{s_1^2/n_1+s_2^2/n_2}$	$H_0:\mu_1-\mu_2=d_0;\ H_1:\mu_1-\mu_2\neq d_0$ $H_0:\mu_1-\mu_2\leq d_0;\ H_1:\mu_1-\mu_2>d_0$ $H_0:\mu_1-\mu_2\geq d_0;\ H_1:\mu_1-\mu_2<d_0$	$t_c=\dfrac{\left(\bar{x}_1-\bar{x}_2\right)-d_0}{\sqrt{s_1^2/n_1+s_2^2/n_2}}\sim t_f$	$\left\{\left\lvert t_c\right\rvert>t_{f,\alpha/2}\right\}$ $\left\{t_c>t_{f,\alpha}\right\}$ $\left\{t_c<-t_{f,\alpha}\right\}$
9	Para $\mu_1-\mu_2$ caso general (n_1 y n_2 grandes)	$\left(\bar{x}-\bar{y}\right)\pm z_{\alpha/2}\sqrt{s_1^2/n_1+s_2^2/n_2}$	$H_0:\mu_1-\mu_2=d_0;\ H_1:\mu_1-\mu_2\neq d_0$ $H_0:\mu_1-\mu_2\leq d_0;\ H_1:\mu_1-\mu_2>d_0$ $H_0:\mu_1-\mu_2\geq d_0;\ H_1:\mu_1-\mu_2<d_0$	$z_c=\dfrac{\left(\bar{x}_1-\bar{x}_2\right)-d_0}{\sqrt{s_1^2/n_1+s_2^2/n_2}}\sim N\left(0,1\right)$	$\left\{\left\lvert z_c\right\rvert>z_{\alpha/2}\right\}$ $\left\{z_c>z_\alpha\right\}$ $\left\{z_c<-z_\alpha\right\}$
10	Para p_1-p_2 caso general (n_1 y n_2 grandes)	$\left(\hat{p}_1-\hat{p}_2\right)\pm z_{\alpha/2}\sqrt{\hat{p}_1\hat{q}_1/n_1+\hat{p}_2\hat{q}_2/n_2}$	$H_0:p_1-p_2=p_0;\ H_1:p_1-p_2\neq p_0$ $H_0:p_1-p_2\leq p_0;\ H_1:p_1-p_2>p_0$ $H_0:p_1-p_2\geq p_0;\ H_1:p_1-p_2<p_0$	$z_c=\dfrac{\left(\hat{p}_1-\hat{p}_2\right)-p_0}{\sqrt{\hat{p}_1\hat{q}_1/n_1+\hat{p}_2\hat{q}_2/n_2}}\sim N\left(0,1\right)$	$\left\{\left\lvert z_c\right\rvert>z_{\alpha/2}\right\}$ $\left\{z_c>z_\alpha\right\}$ $\left\{z_c<-z_\alpha\right\}$
11	Para σ_1^2/σ_2^2 de $N\left(\mu_1,\sigma_1^2\right)$ y $N\left(\mu_2,\sigma_2^2\right)$	$\left(\dfrac{s_1^2/s_2^2}{F_{n_1-1,n_2-1,\alpha/2}},\dfrac{s_1^2}{s_2^2}F_{n_2-1,n_1-1,\alpha/2}\right)$	$H_0:\sigma_1^2=\sigma_2^2;\ H_1:\sigma_1^2\neq\sigma_2^2$ $H_0:\sigma_1^2\leq\sigma_2^2;\ H_1:\sigma_1^2>\sigma_2^2$ $H_0:\sigma_1^2\geq\sigma_2^2;\ H_1:\sigma_1^2<\sigma_2^2$	$F_c=\dfrac{s_1^2}{s_2^2}\sim F_{n_1-1,n_2-1}$	$\left\{F_c\notin\left[F_{n_1-1,n_2-1,1-\alpha/2},F_{n_1-1,n_2-1,\alpha/2}\right]\right\}$ $\left\{F_c>F_{n_1-1,n_2-1,\alpha}\right\}$ $\left\{F_c<F_{n_1-1,n_2-1,1-\alpha}\right\}$